

Assignment

Q show that 2 is a primitive root modulo 11.

→ A number g is a primitive root modulo p (prime p) if its order modulo p is $\phi(p) = p-1$. For $p=11$ we need the order of 2 modulo 11 to be 10.

compute powers of 2 mod 11.

$$2^1 = 2 \quad 2^9 = 6$$

$$2^2 = 4 \quad 2^{10} = 12 = 1$$

$$2^3 = 8$$

$$2^4 = 16 = 5$$

$$2^5 = 10$$

$$2^6 = 20 = 9$$

$$2^7 = 18 = 7$$

$$2^8 = 14 = 3$$

Thus the order of 2 modulo 11 is 10, so 2 is a primitive root modulo 11.

How many incongruent primitive roots does 14 have?

→ if primitive roots exist for n , their number is $\phi(\phi(n))$. primitive roots exist for $n = 2, 4, p^k, 2p^k$ where p is an odd prime. Here $14 = 2 \cdot 7 = 2p$ with $p = 7$, so primitive roots do exist.

$$\text{compute: } \phi(14) = \phi(2) \phi(7) = 1 \cdot 6$$

$$\phi(\phi(14)) = \phi(6) = 2$$

so there are 2 incongruent primitive roots modulo 14.

Q) suppose n is a positive integer, and a^{-1} is a multiplicative inverse of $a \pmod n$.

(a) show $\text{ord}_n a = \text{ord}_n(a^{-1})$

(b) If a is a primitive root modulo n , must a^{-1} also be a primitive root?

soln:

(a.) let $\text{ord}_n(a) = d$. By definition $a^d \equiv 1 \pmod n$ and d is the least positive integer with that property. then

$$(a^{-1})^d = a^{-d} \equiv 1 \pmod n,$$

so the order of a^{-1} divides d : ~~$\text{ord}_n(a^{-1}) \mid d$ and $d \mid \text{ord}_n(a^{-1})$~~ conversely,

if $\text{ord}_n(a^{-1}) = e$, then

$$a^e = (a^{-1})^e)^{-1} \equiv 1^{-1} \equiv 1 \pmod n,$$

so $d \mid e$. combining the two divisibility

$\text{ord}_n(a^{-1}) \mid d$ and $d \mid \text{ord}_n(a^{-1})$ gives
 $\text{ord}_n(a^{-1}) = d$, then

$$ae = ((a^{-1})e)^{-1} \equiv 1^{-1} \equiv 1 \pmod{n},$$

so by combining the two
divisibilities $\text{ord}_n(a^{-1}) \mid d$ and $d \mid \text{ord}_n(a^{-1})$ gives,

$$\text{ord}_n(a^{-1}) = d = \text{ord}_n(a).$$

(b:) If a is a primitive root mod
 n , then $\text{ord}_n(a) = \varphi(n)$. From part (a)
we have $\text{ord}_n(a^{-1}) = \text{ord}_n(a)$. Hence
 $\text{ord}_n(a^{-1}) = \varphi(n)$, so a^{-1} is also a
primitive root modulo n .